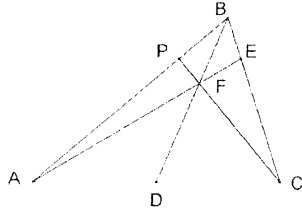


JUNIOR

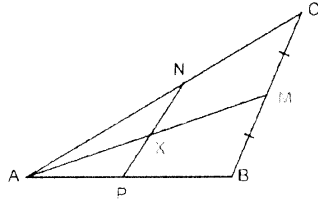
1. In $\triangle ABC$, D is the midpoint of AC and E is a point on BC such that $\frac{BC}{BE} = 4$. Let BD and AE meet in F. If area of $\triangle ABF = 1 \text{ cm}^2$, then area of $\triangle ABC$ in cm^2 is



- (1) 3
(2) 4
(3) 5
(4) 6
2. Let x and y be real numbers such that $6x^2 + 2xy + 6y^2 = 9$. Then the maximum value of $x^2 + y^2$ is
- (1) 5
(2) 1
(3) $\frac{7}{5}$
(4) $\frac{9}{5}$
3. A *cascading number* is one where the digits are strictly decreasing from left to right, like 54321 or 65420. Among all 5 digit cascading numbers from 10000 to 99999 the 202nd number does not have digit
- (1) 4
(2) 5
(3) 8
(4) 9
4. A function f from integers to integers is defined as
- $$f(n) = \begin{cases} n + 5 & \text{if } n \text{ is odd} \\ n/2 & \text{if } n \equiv 2 \pmod{4} \text{ and} \\ n/4 & \text{if } n \equiv 0 \pmod{4}. \end{cases}$$
- If $f(f(f(n))) = 19$, then the number of such n is
- (1) 2
(2) 3
(3) 4
(4) more than four
5. If a fraction F is written as $0.282828\dots$ in base A and as $0.33333\dots$ in base B then least value of $A + B$ is
- (1) 17
(2) 20

- (3) 22
(4) 24

6. In $\triangle ABC$, M is the midpoint of BC. AB = 20 cm and AC = 30 cm. N and P, are points on AC and AB respectively such that AN = 3AP. NP and AM meet at X. Then $\frac{NX}{XP}$ equals



- (1) 2
(2) $\frac{3}{4}$
(3) $\frac{1}{2}$
(4) $\frac{4}{3}$

7. Let $f(x) = A \sin^3 x + B \tan x + C x^5 - 2$. Given that $f(\log(\log_4 100)) = 5$. Then $f(\log(\log 2))$ equals

- (1) -5
(2) 7
(3) -7
(4) -9

8. How many three digit positive integers are there such that each is a multiple of 11 that contains distinct digits?

- (1) 24
(2) 32
(3) 44
(4) none of these

9. How many prime numbers p, are there such that $16p - 27$ is the cube of a positive integer?

- (1) 0
(2) 1
(3) 2
(4) 3

10. A, B, C are positive integers such that $A \cdot \log_{324}(27) + B \cdot \log_{324}(8) = C$. The least value of $A + B + C$ is

- (1) 5
(2) 7
(3) 9
(4) 11

11. ABCDE is a concave pentagon that can be dissected into exactly two equilateral triangles each of side length greater than one cm. The area of the pentagon is $5\sqrt{3}$ cm² and its perimeter is 14 cm. The sum of the lengths of adjacent sides containing the largest vertex angle of the pentagon is

(1) 4 cm
(2) 6 cm
(3) 3 cm
(4) 4.4 cm

12. a, b, c are integers such that $ax^2 + bx + c$ is exactly divisible by 5 for every integer x. Then, which of the following is a possible remainder when abc is divided by 1000?

(1) 150
(2) 200
(3) 250
(4) 300

13. What is the number of subsets of {1, 2, 3, 4, 5, 6, 7, 8} that are subsets of neither {1, 2, 3, 4, 5} nor {4, 5, 6, 7, 8}?

(1) 196
(2) 192
(3) 128
(4) 188

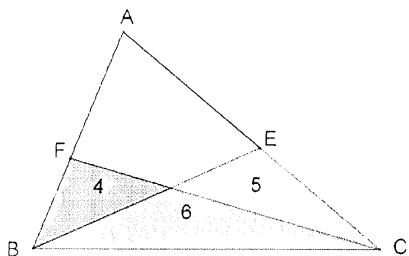
14. Let $N = 123456\dots 41424344$ be the 79 digit number formed by writing 1, 2, 3, ..., 43, 44 in order one after the other. The remainder when N is divided by 45 is

(1) 4
(2) 9
(3) 18
(4) 27

15. If $f(x) = x^4 + ax^3 + bx^2 + cx + d$ is a polynomial such that $f(1) = f(3) = f(4) = f(6)$, what is the value of b?

(1) 60
(2) -60
(3) 67
(4) -67

16. In the following figure the areas of the shaded regions are given in cm². What is the area of the quadrilateral left out in cm²?

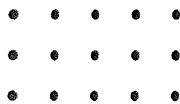


- (1) 25.5
- (2) 26.75
- (3) 26.25
- (4) 27

17. The function $f(x)$ is defined for all real numbers x . If $f(a + b) = f(ab)$ for all $a, b \in \mathbb{R}$ and $f(\frac{1}{2}) = \frac{1}{2}$, then $f(2020)$ is equal to

- (1) 2020
- (2) $-\frac{1}{2}$
- (3) 0
- (4) $\frac{1}{2}$

18. A 3×5 grid of dots is given as shown. How many straight line segments can be drawn that join two of those dots and pass through exactly one other dot?



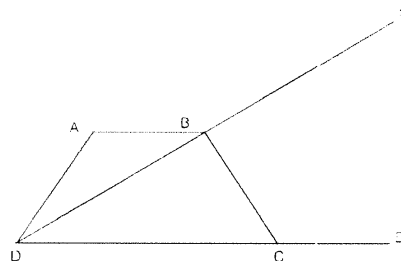
- (1) 14
- (2) 16
- (3) 20
- (4) 22

19. Professor Absentminded has a very peculiar problem, in that he cannot remember numbers larger than 15. However, he tells his wife, "I can remember any number up to 100 by remembering the three numbers obtained as remainders when the number is divided by 5, 7 and 11, respectively. For example (2, 3, 6) is 17". Professor remembers that he had (0, 0, 4) rupees in the purse, and he paid (3, 2, 1) rupees to the servant. How much money is left in the purse in Rs.?

- (1) 47
- (2) 36
- (3) 25
- (4) None of these

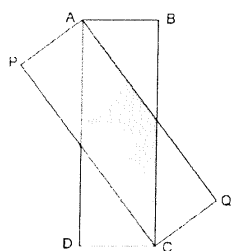
20. ABCD is an isosceles trapezium with $AD = BC = 10$, $AB = 6$ and $DC = 22$. E lies on

DC extended and FE is perpendicular to DE with D, B, F collinear and B the midpoint of DF. Then CE in cm =



- (1) 4
- (2) 5
- (3) 6
- (4) 7

21. Two identical rectangles of size 1 cm x 3 cm are overlapping so that a diagonal of one falls on the diagonal of the other. If S is the common shaded area between the two, then S equals in cm^2



- (1) $5/2$
- (2) $5/3$
- (3) $5/4$
- (4) $4/3$

22. Let x, y, z be real numbers such that $\frac{xy}{x+y} = \frac{1}{3}, \frac{yz}{y+z} = \frac{1}{5}, \frac{zx}{z+x} = \frac{1}{6}$, then the value of $\frac{7xyz}{xy+yz+zx}$ is

- (1) 7
- (2) 2
- (3) 1
- (4) 0.5

23. x, y are positive integers such that $101x + 17y = 2020$. The value of $x + y =$

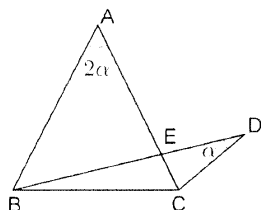
- (1) 20
- (2) 104
- (3) 101
- (4) No solution exists

24. An ordered 6 tuple of positive integers $(a_1, a_2, a_3, a_4, a_5, a_6)$ satisfies both the following properties: $a_n + a_{n+1} = a_{n+2}$ for $1 \leq n \leq 4, n \in \mathbb{N}$ and $a_5 = 2020$. How many such ordered 6

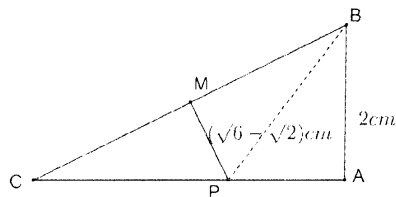
tuples of positive integers are there ?

- (1) 337
- (2) 135
- (3) 336
- (4) 134

25. A is a point in the plane of $\triangle BCD$ such that $AB = AC$. Also, $\angle BAC = 2\angle BDC$. Also, AC and BD meet at E. If $AB = 6$ and $EC = 2$ units, then BE. ED = ----- unit²



- (1) 24
 - (2) 20
 - (3) 8
 - (4) None of these
26. ABC is a right triangle where $\angle BAC = 90^\circ$. M, P, are points on sides BC, AC respectively such that MP is the perpendicular bisector of BC, as shown. If $AB = 2$ cm and $MP = (\sqrt{6} - \sqrt{2})$ cm, then CP =



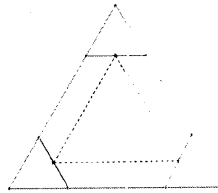
- (1) 4 cm
 - (2) $(\sqrt{6} - \sqrt{2})$ cm
 - (3) $(2 + \sqrt{3})$ cm
 - (4) 3 cm
27. Let x and y be real numbers satisfying $2x^2 - 42xy + 441(y + 1)(y - 1) \leq 42x - 882$. Then the value of $4(x^2 + 64y^2)$ is
- (1) 2020
 - (2) 2021
 - (3) 2019
 - (4) 1729
28. k is a 4-digit number such that the difference between k and its reversed four digit number is 2088. The number of such 4-digit numbers k, is

- (1) 81
- (2) 63
- (3) 126
- (4) 162

29. The picture below shows an equilateral triangle whose sides have length 2020 cm. In each

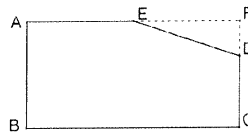
corner of the triangle a smaller equilateral triangle with sides of length 840 cm is inscribed. The midpoints of the inner sides of these three smaller triangles are joined to form an equilateral triangle in the centre of the large equilateral triangle. What is the side

length of this new equilateral triangle in cm?



- (1) 760
- (2) 840
- (3) 680
- (4) None of these

30. A pentagon is formed by cutting off a triangular piece from the corner of a rectangle. The sides of the pentagon in some order are 5, 25, 13, 40 and 37. Then the area of the rectangle is



- (1) 1240
- (2) 1480
- (3) 925
- (4) 1000